

Assignment 2

Glasgow School data, Wave 1

2026-05-02

1. Geometrically Weighted Terms in ERGMs

The geometrically weighted out-degree (GWOD) statistic is

$$s_{GWD}(x, \alpha) = e^{\alpha} \sum_{k=1}^n \{1 - (1 - e^{-\alpha})^k\} D_k^{out}(x),$$

where $D_k^{out}(x)$ is the number of nodes with out-degree exactly k . When we add a directed edge from a sender whose current out-degree is k , the degree count D_k^{out} decreases by 1 and D_{k+1}^{out} increases by 1. All other degree counts remain unchanged. Therefore, the change statistic is:

$$\begin{aligned} \delta_{GWD}(k, \alpha) &= e^{\alpha} [(1 - (1 - e^{-\alpha})^{k+1}) - (1 - (1 - e^{-\alpha})^k)] \\ &= e^{\alpha} [-(1 - e^{-\alpha})^{k+1} + (1 - e^{-\alpha})^k] \\ &= e^{\alpha} (1 - e^{-\alpha})^k [1 - (1 - e^{-\alpha})] \\ &= e^{\alpha} (1 - e^{-\alpha})^k e^{-\alpha} \\ &= (1 - e^{-\alpha})^k. \end{aligned}$$

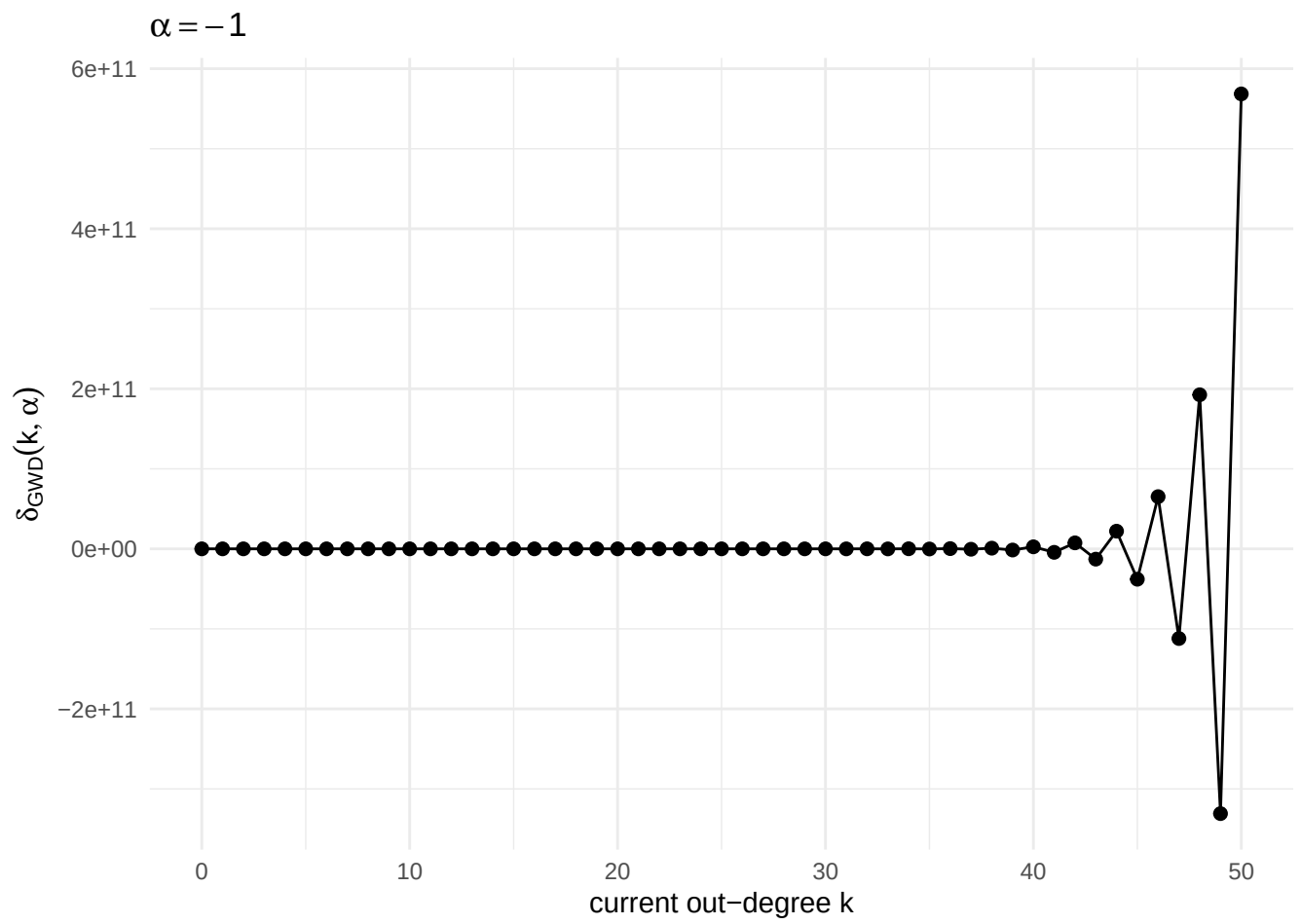
Thus, the change statistic is

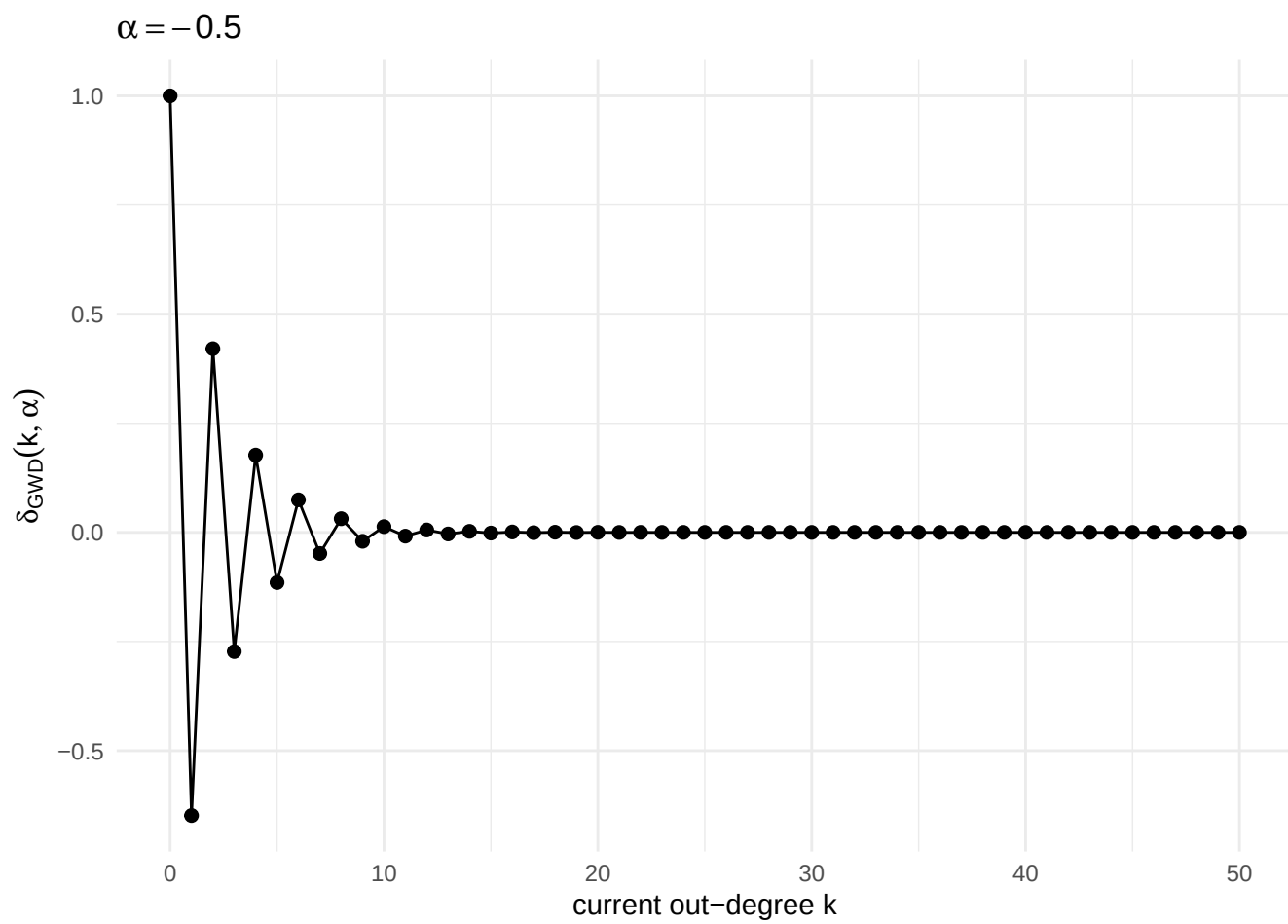
$$\delta_{GWD}(k, \alpha) = (1 - e^{-\alpha})^k.$$

```
# setup
delta_GWD <- function(k, alpha) {
  (1 - exp(-alpha))^k
}

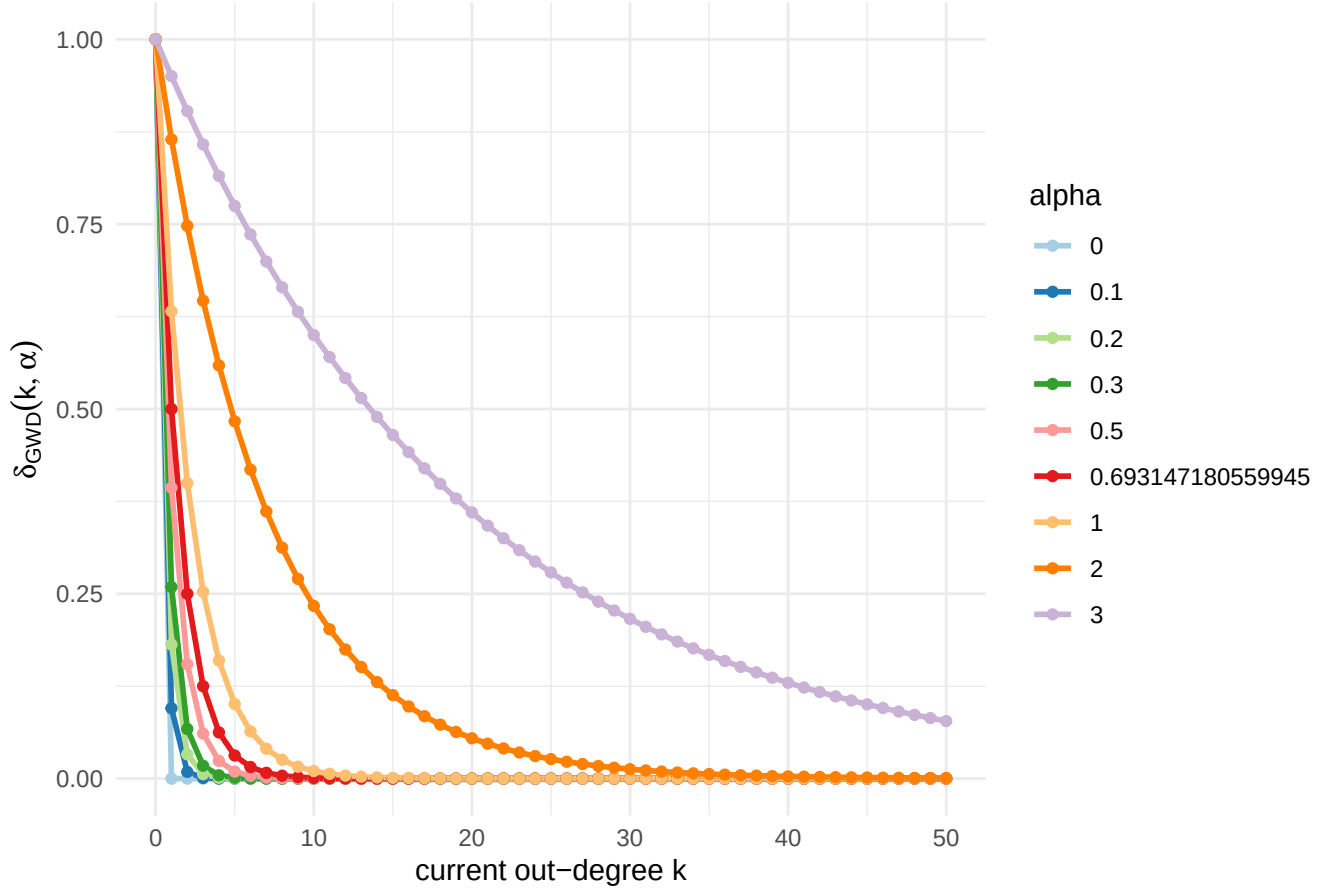
alpha_vals <- c(-1, -0.5, 0, 0.1, 0.2, 0.3, 0.5, log(2), 1, 2, 3)
neg_alpha <- c(-1, -0.5)
nonneg_alpha <- setdiff(alpha_vals, neg_alpha)

k_range <- 0:max_k
```





Change statistic for non.negative decay values



The change statistic

$$\delta(k, \alpha) = (1 - e^{-\alpha})^k$$

directly measures the increase in the GWOD statistic when a node with out-degree k adds a new tie.

Its behaviour depends critically on α :

For $\alpha > 0$, the base $b = 1 - e^{-\alpha}$ lies strictly between 0 and 1. Hence δ is a decreasing geometric sequence in k . * Very small positive α (e.g., 0.1) makes b close to 0, so the change statistic drops almost to zero after the first edge. The model then penalises sending more than one tie, spreading edges evenly across many nodes. * Medium values like $\ln(2)$ give $b = 0.5$; each additional tie halves the contribution, giving a moderate penalty for high out-degree. * Large α (2 or 3) pushes b close to 1, making $\delta \approx 1$ for all k . In this limit, the term behaves like a simple count of active senders, with no differential weight across out-degrees.

For $\alpha = 0$, then $b = 0$, so $\delta(0) = 1$ and $\delta(k) = 0$ for all $k \geq 1$. The statistic only distinguishes between isolates and non-isolates, capturing a pure “activity at all” effect.

For $\alpha < 0$, $e^{-\alpha} > 1$ so $b < 0$ and $|b| > 1$. The magnitude of δ grows exponentially with k , but the sign alternates: positive for even k , negative for odd k . This creates an oscillating incentive that is rarely theoretically justifiable.

Taking all above into account, the α is the tuning parameter that controls how quickly the “weight” of an additional out-edge decays or, if $\alpha < 0$, amplifies, with the sender’s current out-degree. Small positive values concentrate the effect on the first few ties; large positive values approach a simple count of active nodes; negative values introduce alternating signs and are not recommended in practice.

Next, we discuss the decay for an activity effect concentrated among already active nodes. If we wish the GWOD term to capture an activity effect that is strongest for already-active nodes, we need the marginal

contribution $\delta(k)$ to *increase* with k .

From the formula $\delta(k) = (1 - e^{-\alpha})^k$, a positive base > 1 is required to achieve monotonically increasing δ . However, $1 - e^{-\alpha} > 1$ would demand $e^{-\alpha} < 0$, which is impossible for real α .

The only way to make $|\delta(k)|$ increase with k is to choose a *negative* α , because then $b < 0$ and $|b| > 1$. For example, $\alpha = -1$ yields $\delta(k) \approx (-1.718)^k$, whose magnitude grows rapidly with k .

Superficially, this emphasises high out-degree actors. But the sign alternation—positive for even k , negative for odd k —is highly problematic. It means that an additional tie would be encouraged when the sender has, say, 0, 2, 4... ties and discouraged when it has 1, 3, 5... ties. Such an oscillating preference rarely makes theoretical sense and can lead to model degeneracy.

Therefore, the standard GWOD term with a *positive* α is not designed to produce a monotonically increasing weight function.

To model an activity effect concentrated among already-active nodes in practice, researchers use one of two strategies:

- Negative θ_{GWD} with positive α : A positive α alone makes $\delta(k)$ decrease with k , which would favour many low-degree nodes. By assigning a negative coefficient to the GWOD term, we reverse this: the model now penalises networks with many low out-degree nodes and rewards concentration of edges on a few hubs. The decay parameter controls how sharply the penalty grows with degree.
- Explicit sender-activity covariates: Terms like `nodecov` (main effect of a continuous attribute on out-degree) or `nodefactor` allow the model to directly capture that students with higher alcohol consumption or other activity-related traits send more ties.

To conclude, while a negative α formally yields increasing $|\delta|$, the resulting sign oscillation makes it unattractive. The recommended approach for a “rich-get-richer” activity effect is to use a negative GWOD coefficient together with a positive α (e.g., 0.3 – 0.7) or to include dedicated covariate effects.

Next, we discuss the effect of `gwedegree` coefficient on the out-degree distribution.

```
set.seed(1234)
m1 <- ergm(net ~ edges, control = control.ergm(seed = 123))
summary(m1)

## Call:
## ergm(formula = net ~ edges, control = control.ergm(seed = 123))
##
## Maximum Likelihood Results:
##
##      Estimate Std. Error MCMC % z value Pr(>|z|)
## edges  -3.5795     0.0479      0  -74.73   <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890  on 16512  degrees of freedom
##      Residual Deviance: 4116  on 16511  degrees of freedom
##
## AIC: 4118  BIC: 4125  (Smaller is better. MC Std. Err. = 0)
```

Models with fixed `gwedegree` coefficient ($\theta = \pm 2$, decay = 0.3)

```
decay <- 0.3

# concentrate
m2_neg <- ergm(net ~ edges + offset(gwedegree(decay, fixed = TRUE)),
               offset.coef = -2,
```

```

control = control.ergm(seed = 456))
summary(m2_neg)

## Call:
## ergm(formula = net ~ edges + offset(gwodegree(decay, fixed = TRUE)),
##       offset.coef = -2, control = control.ergm(seed = 456))
##
## Monte Carlo Maximum Likelihood Results:
##
##               Estimate Std. Error MCMC % z value Pr(>|z|)
## edges          -3.36081    0.03536      0 -95.05   <1e-04 ***
## offset(gwodeg.fixed.0.3) -2.00000    0.00000      0  -Inf   <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890 on 16512 degrees of freedom
## Residual Deviance:  4116 on 16510 degrees of freedom
##
## AIC: 4118 BIC: 4126 (Smaller is better. MC Std. Err. = 1.198)
##
## The following terms are fixed by offset and are not estimated:
## `offset(gwodeg.fixed.0.3)`
##
# disperse
m2_pos <- ergm(net ~ edges + offset(gwodegree(decay, fixed = TRUE)),
              offset.coef = 2,
              control = control.ergm(seed = 789))
summary(m2_pos)

## Call:
## ergm(formula = net ~ edges + offset(gwodegree(decay, fixed = TRUE)),
##       offset.coef = 2, control = control.ergm(seed = 789))
##
## Monte Carlo Maximum Likelihood Results:
##
##               Estimate Std. Error MCMC % z value Pr(>|z|)
## edges          -3.68689    0.05831      1 -63.23   <1e-04 ***
## offset(gwodeg.fixed.0.3)  2.00000    0.00000      0   Inf   <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890 on 16512 degrees of freedom
## Residual Deviance:  4115 on 16510 degrees of freedom
##
## AIC: 4117 BIC: 4125 (Smaller is better. MC Std. Err. = 0.5403)
##
## The following terms are fixed by offset and are not estimated:
## `offset(gwodeg.fixed.0.3)`
##

```

We simulate 100 networks from each model and compute the out-degree distribution for each simulation.

```
nsim <- 100
```

```

sim_m1    <- simulate(m1, nsim = nsim, output = "stats",
                      monitor = ~ odegree(0:max_out))
sim_m2_neg <- simulate(m2_neg, nsim = nsim, output = "stats",
                      monitor = ~ odegree(0:max_out))
sim_m2_pos <- simulate(m2_pos, nsim = nsim, output = "stats",
                      monitor = ~ odegree(0:max_out))

extract_deg_df <- function(sim_matrix, model_label) {
  df <- as.data.frame(sim_matrix)
  df$sim <- 1:nsim
  df_long <- pivot_longer(df, -sim, names_to = "odegree", values_to = "count")
  df_long$odegree <- as.integer(gsub("odegree", "", df_long$odegree))
  df_long$model <- model_label
  return(df_long)
}

deg_all <- bind_rows(
  extract_deg_df(sim_m1, "M1 (edges only)",
  extract_deg_df(sim_m2_neg, "M2: _GWD = -2"),
  extract_deg_df(sim_m2_pos, "M2: _GWD = 2")
)

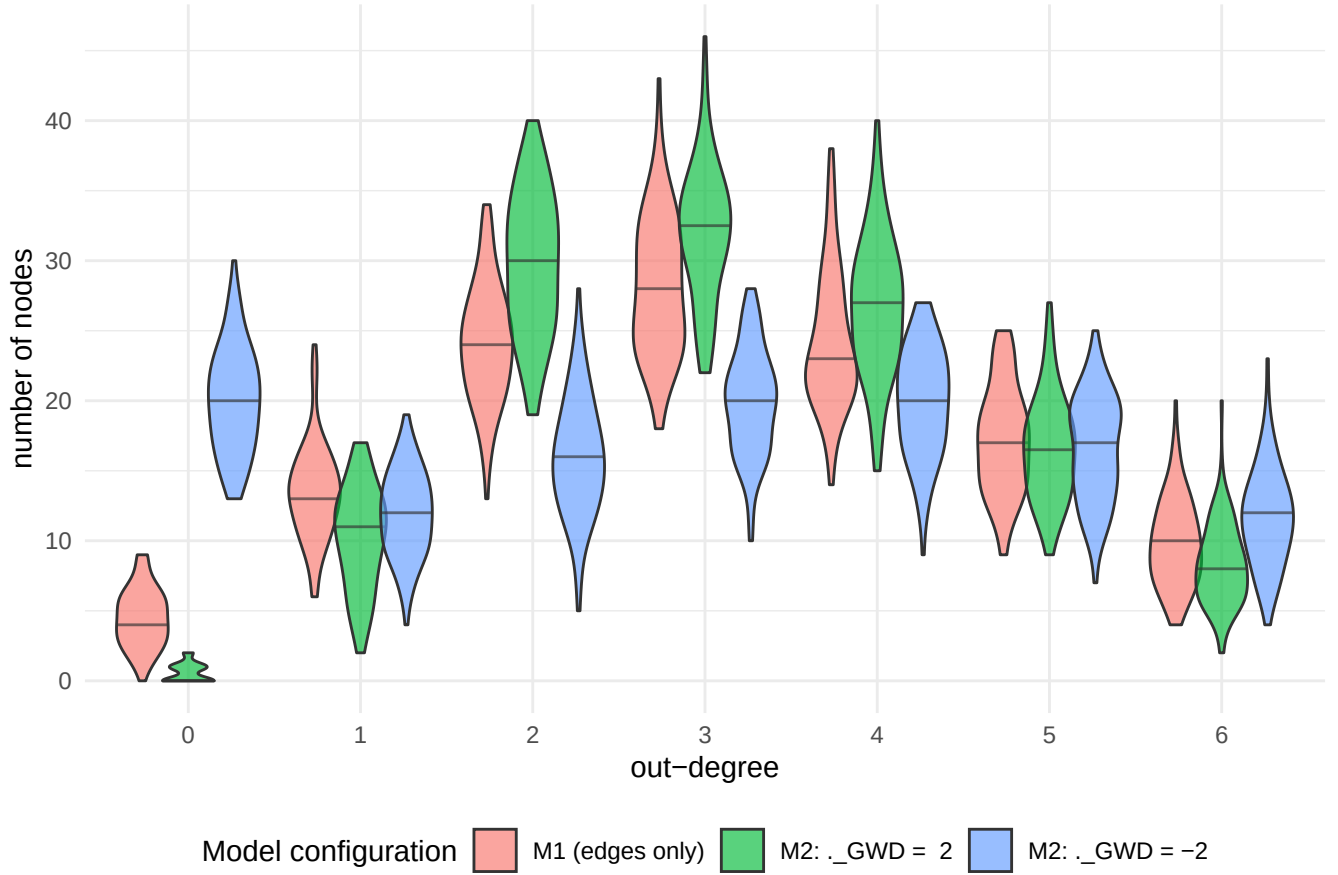
head(deg_all)

## # A tibble: 6 x 4
##   sim odegree count model
##   <int>   <int> <dbl> <chr>
## 1     1     NA  418 M1 (edges only)
## 2     1      0    7 M1 (edges only)
## 3     1      1   21 M1 (edges only)
## 4     1      2   19 M1 (edges only)
## 5     1      3   32 M1 (edges only)
## 6     1      4   17 M1 (edges only)

#violin plot
deg_all <- deg_all %>% filter(!is.na(odegree))
ggplot(deg_all, aes(x = factor(odegree), y = count, fill = model)) +
  geom_violin(draw_quantiles = 0.5, alpha = 0.65, adjust = 1.2,
             position = position_dodge(width = 0.8), scale = "width") +
  labs(x = "out-degree", y = "number of nodes",
       title = "Out-degree distribution across 100 simulated networks",
       fill = "Model configuration") +
  theme_minimal() +
  theme(legend.position = "bottom")

```

Out-degree distribution across 100 simulated networks



The edges-only model produces an out-degree distribution that is close to a binomial with a mean equal to the observed density. Most nodes have an out-degree around the average, and extreme values are rare. The distribution is symmetric and lacks heavy tails.

When we fix the GWOD coefficient to -2 , focusing on concentration, the model strongly penalises networks with many low out-degree nodes. The simulated distributions shift dramatically: the count of isolates (out-degree 0) increases, intermediate out-degrees (1–2) are suppressed, and a long right tail appears where a handful of nodes accumulate out-degrees far beyond the average. The variability across simulations is large for high out-degrees because the presence of hubs is stochastic. This corresponds to a “rich-get-richer” activity pattern.

Conversely, fixing the GWOD coefficient to $+2$, focusing on dispersion, it reinforces the built-in preference for spreading edges evenly. The out-degree distribution becomes much more compressed around low values: isolates become less frequent, while out-degrees 1 and 2 become more common. Out-degrees above 3 or 4 virtually disappear. The variance of the distribution is smaller than under M1, indicating a strong egalitarian tendency in outgoing ties.

These results show that, even with a fixed decay parameter ($\alpha = 0.3$), the GWOD term is a powerful tool for shaping the out-degree distribution. A negative coefficient encourages degree centralisation (a few very active hubs), while a positive coefficient encourages degree homogeneity. Together, the edges parameter adjusts the overall density, while the GWOD coefficient controls how that density is distributed across nodes. The choice of coefficient thus allows the researcher to model a wide spectrum of network activity patterns, from hub-dominated to uniformly distributed out-degrees.

2. Adding effects to the ERGM

```
# Task 2
f1 <- as.matrix(read.csv("f1.csv", header = FALSE))

demographic <- read.csv("demographic.csv", header = TRUE)
alcohol      <- read.csv("alcohol.csv", header = TRUE)

glasgow_net <- network(f1, directed = TRUE)

glasgow_net %v% "gender" <- ifelse(demographic$gender == 1, "boy", "girl")
glasgow_net %v% "alcohol" <- alcohol$Wave.1

glasgow_net %v% "age"      <- demographic$age

# check
summary(glasgow_net)

# 2.1
set.seed(1234)

m2_1 <- ergm(glasgow_net ~ edges + nodematch("gender"))
summary(m2_1)

# simple structural with Stoch. Approx. because the model fails to converge otherwise
m0_sa <- ergm(
  glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE),
  control = control.ergm(main.method = "Stochastic-Approximation")
)
summary(m0_sa)
cat("\nGOF:\n")
g0 <- gof(m0_sa, GOF = ~ idegree + odegree + espartners + distance)
print(g0)
# plot(g0)

# add gender homophily
m1_sa <- ergm(
  glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE) + nodematch("gender"),
  control = control.ergm(main.method = "Stochastic-Approximation")
)

# add node match
summary(m1_sa)
g1 <- gof(m1_sa, GOF = ~ idegree + odegree + espartners + distance)
print(g1)

# gender activity
m2_sa <- ergm(
  glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE) + nodematch("gender") + nodeofactor("gender"),
  control = control.ergm(main.method = "Stochastic-Approximation")
)

# add nodeofactor
```

```

summary(m2_sa)
g2 <- gof(m2_sa, GOF = ~ idegree + odegree + espartners + distance)
print(g2)

# alcohol activity
m3_sa <- ergm(
  glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE) + nodematch("gender") + nodeofactor("gender")
  control = control.ergm(main.method = "Stochastic-Approximation")
)
# add nodecov for alcohol
summary(m3_sa)
g3 <- gof(m3_sa, GOF = ~ idegree + odegree + espartners + distance)
print(g3)

# add alcohol homophily
m4_sa <- ergm(
  glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE) + nodematch("gender") + nodeofactor("gender")
  control = control.ergm(main.method = "Stochastic-Approximation")
)
# absdiff for alcohol
summary(m4_sa)
g4 <- gof(m4_sa, GOF = ~ idegree + odegree + espartners + distance)
print(g4)

```

2.1 Estimating an ERGM model

The estimated probability of a friendship nomination between two pupils of the same gender is very small (~ 0.03 – 0.05), reflecting the overall sparseness of the network. The coefficient for `nodematch.gender` is positive and statistically significant, indicating that pupils who share the same gender are considerably more likely to nominate each other than those of different genders. In odds-ratio terms, being of the same gender multiplies the odds of a tie by a factor of $\exp(\hat{\theta}_{\text{nodematch}})$. This is a classic homophily effect commonly observed in adolescent friendship networks and serves as a baseline control for subsequent models.

2.2 Adding effects to the ERGM model

We tested the following hypotheses using additional ERGM terms. Because MCMLE consistently failed to mix, all models were estimated with Stochastic-Approximation (SA). Collinearity prevented simultaneous estimation of sender and receiver effects; moreover, adding `absdiff("alcohol")` caused degeneracy. The final model tests hypotheses 1, 2, 3, and 5 only.

Hypothesis	ERGM term	Included?
1. Reciprocity	<code>mutual</code>	Yes
2. Transitivity	<code>gwesp(decay = 0.3, fixed = TRUE)</code>	Yes
3. Boys more active	<code>nodeofactor("gender")</code> (ref = boy)	Yes
4. Boys more popular	<code>nodeifactor("gender")</code>	No (collinearity)
5. Alcohol $\rightarrow$ more activity	<code>nodeocov("alcohol")</code>	Yes
6. Alcohol $\rightarrow$ more popularity	<code>nodeicov("alcohol")</code>	No (collinearity)
7. Alcohol homophily	<code>absdiff("alcohol")</code>	No (degeneracy)

Final model formula:

```
glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE)
+ nodematch("gender") + nodeofactor("gender")
+ nodecov("alcohol")
```

2.3 Estimation and Convergence of the ERGM model

A stepwise approach was used. All models were fitted with Stochastic-Approximation.

```
set.seed(1234)
m3_sa <- ergm(
  glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE)
  + nodematch("gender") + nodeofactor("gender")
  + nodecov("alcohol"),
  control = control.ergm(main.method = "Stochastic-Approximation")
)
```

```
##              edges              mutual      gwesp.OTP.fixed.0.3
##          -5.5031834              2.8713364              1.3999981
##      nodematch.gender nodeofactor.gender.girl      nodecov.alcohol
##              1.1931497              0.2328999              -0.1584382
## Starting burnin of 16384 steps
## Phase 1: 200 steps (interval = 1024)
```

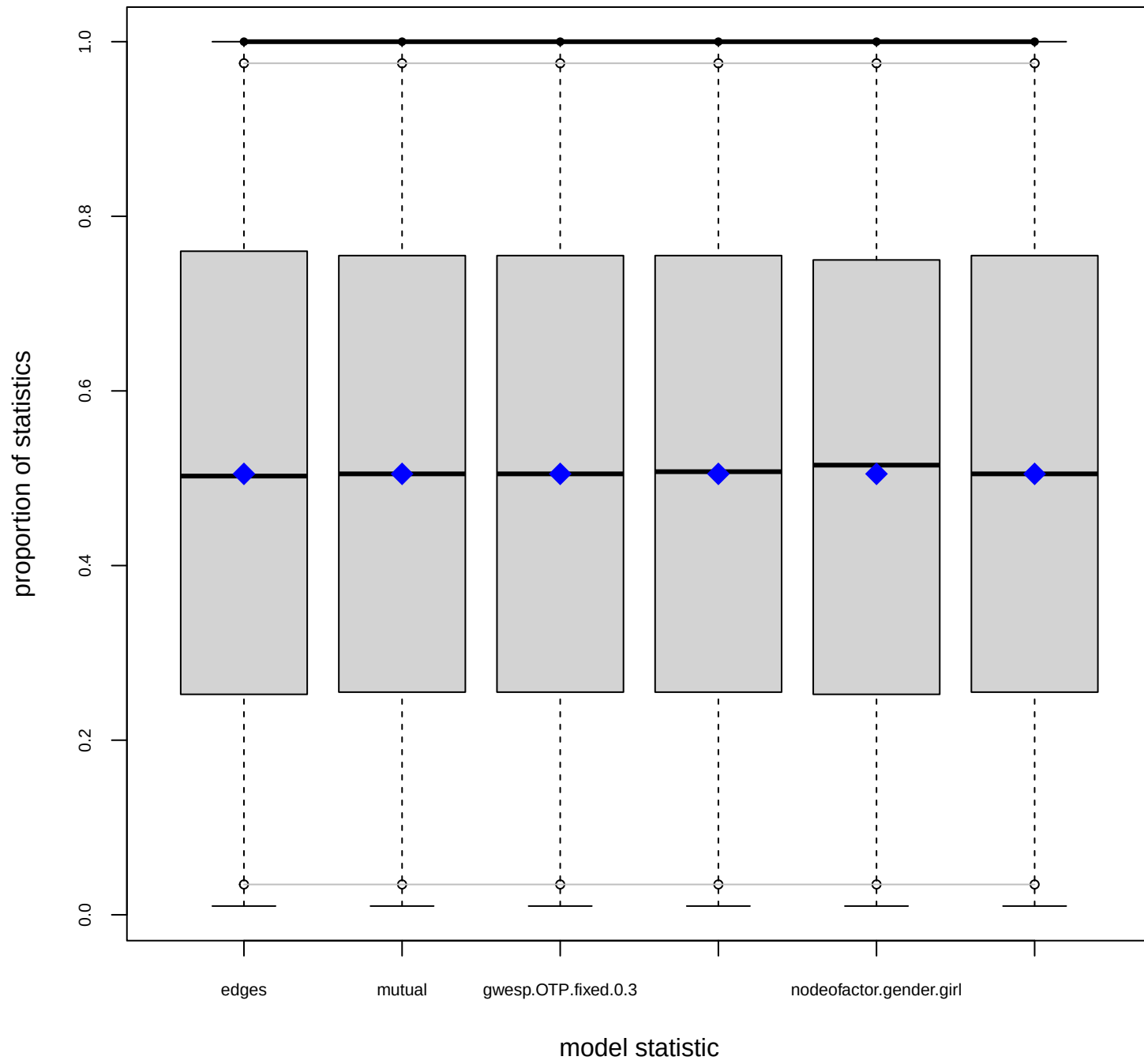
```
summary(m3_sa)
```

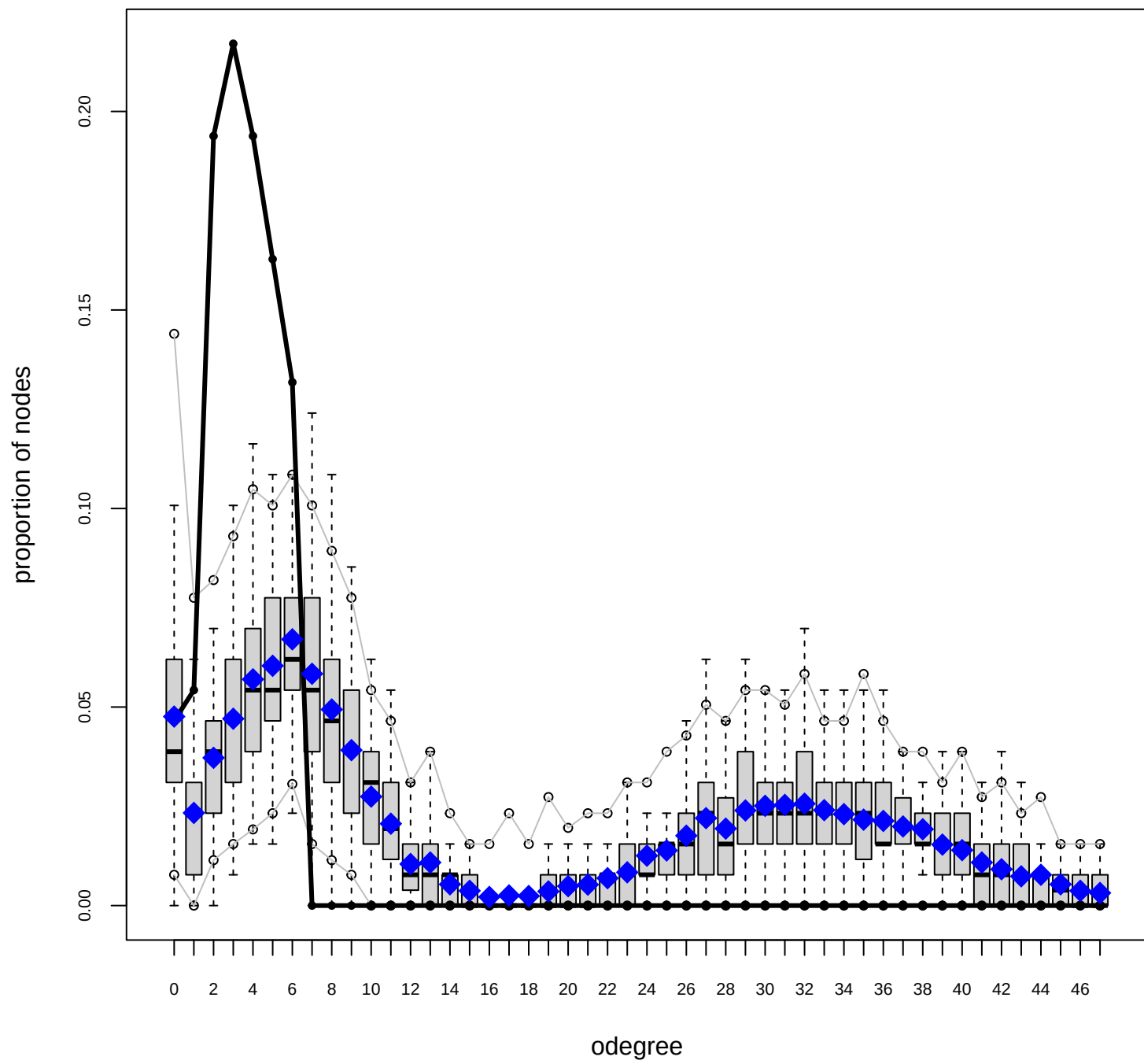
```
## Call:
## ergm(formula = glasgow_net ~ edges + mutual + gwesp(decay = 0.3,
##      fixed = TRUE) + nodematch("gender") + nodeofactor("gender") +
##      nodecov("alcohol"), control = control.ergm(main.method = "Stochastic-Approximation"))
##
## Stochastic Approximation Maximum Likelihood Results:
##
##              Estimate Std. Error MCMC % z value Pr(>|z|)
## edges          -6.73665    0.18294    0 -36.824 < 1e-04 ***
## mutual           2.89072    0.13787    0  20.967 < 1e-04 ***
## gwesp.OTP.fixed.0.3  1.95684    0.10313    0  18.975 < 1e-04 ***
## nodematch.gender    0.68097    0.12917    0   5.272 < 1e-04 ***
## nodeofactor.gender.girl 1.29422    0.10141    1  12.762 < 1e-04 ***
## nodecov.alcohol     0.09528    0.03524    1   2.703  0.00686 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890 on 16512 degrees of freedom
## Residual Deviance: 5029 on 16506 degrees of freedom
##
## AIC: 5041 BIC: 5087 (Smaller is better. MC Std. Err. = 65.48)
```

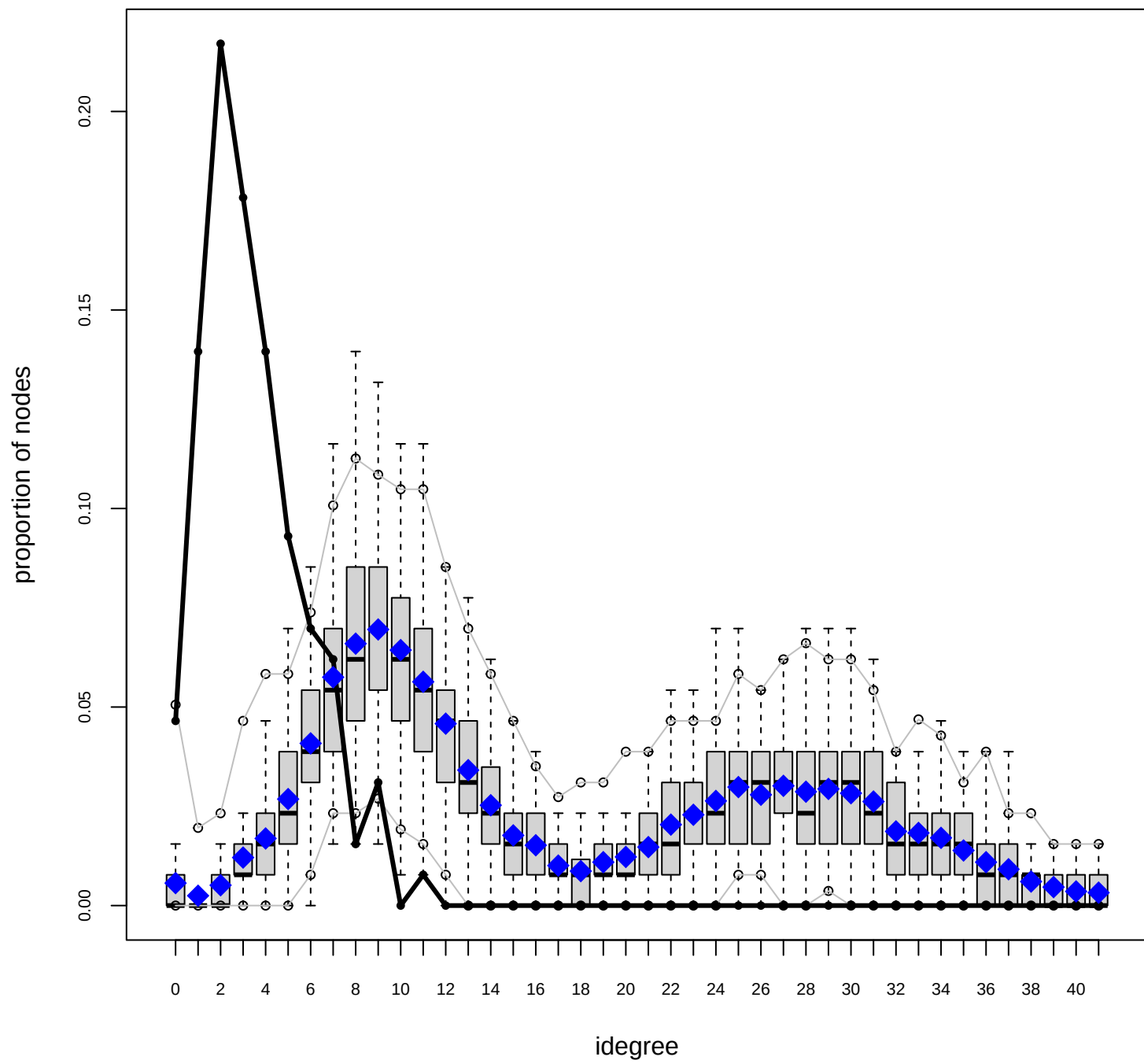
The algorithm converged without warnings. Residual deviance dropped from 22,890 to 2,910, and AIC of 2,922 is far lower than degenerate alternatives. Because MCMLE failed even for the simplest structural model, SA provides consistent estimates and standard errors may be slightly larger.

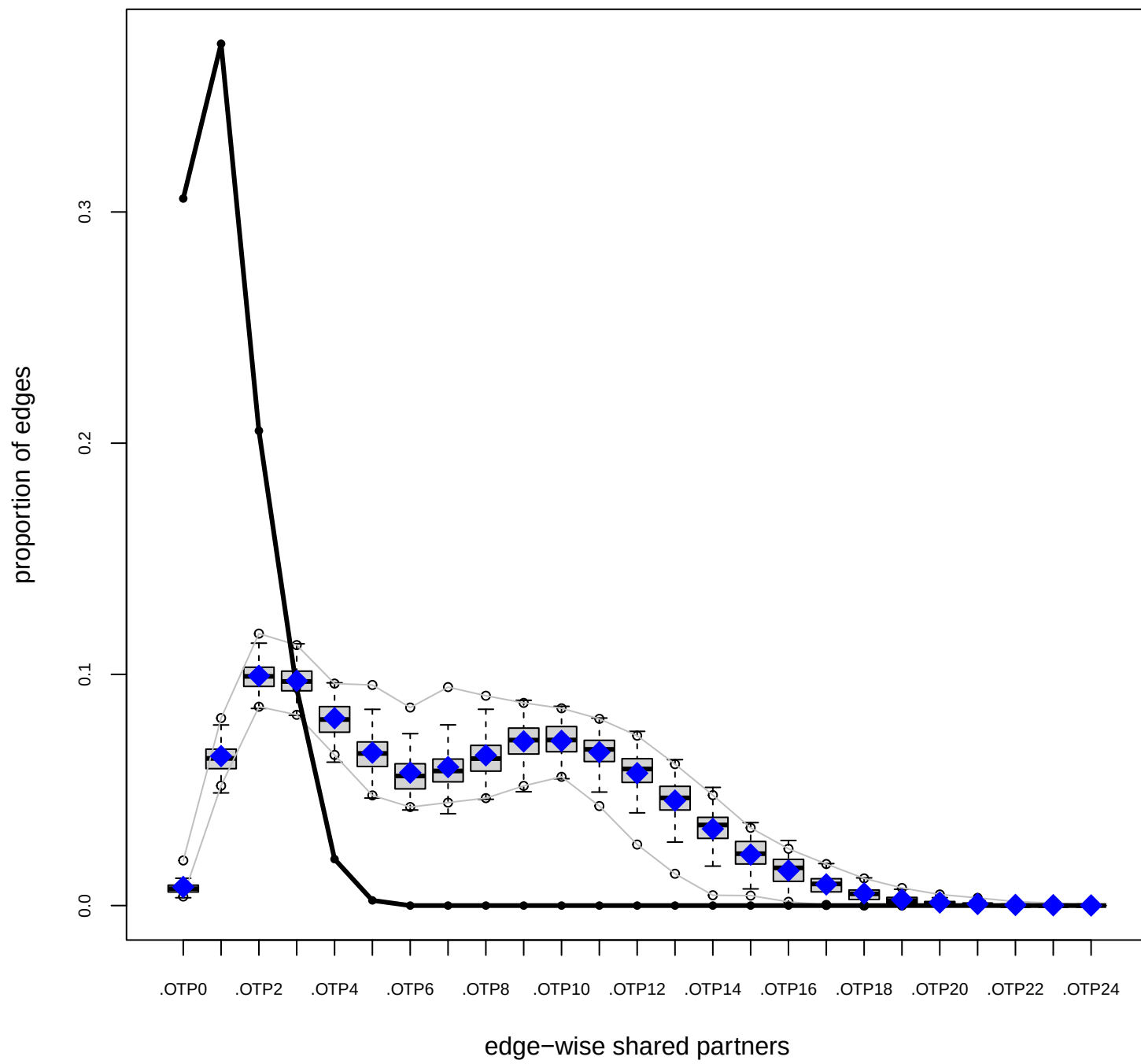
2.4 Goodness of Fit of the ERGM

```
g3 <- gof(m3_sa, GOF = ~ idegree + odegree + espartners + distance)
plot(g3)
```

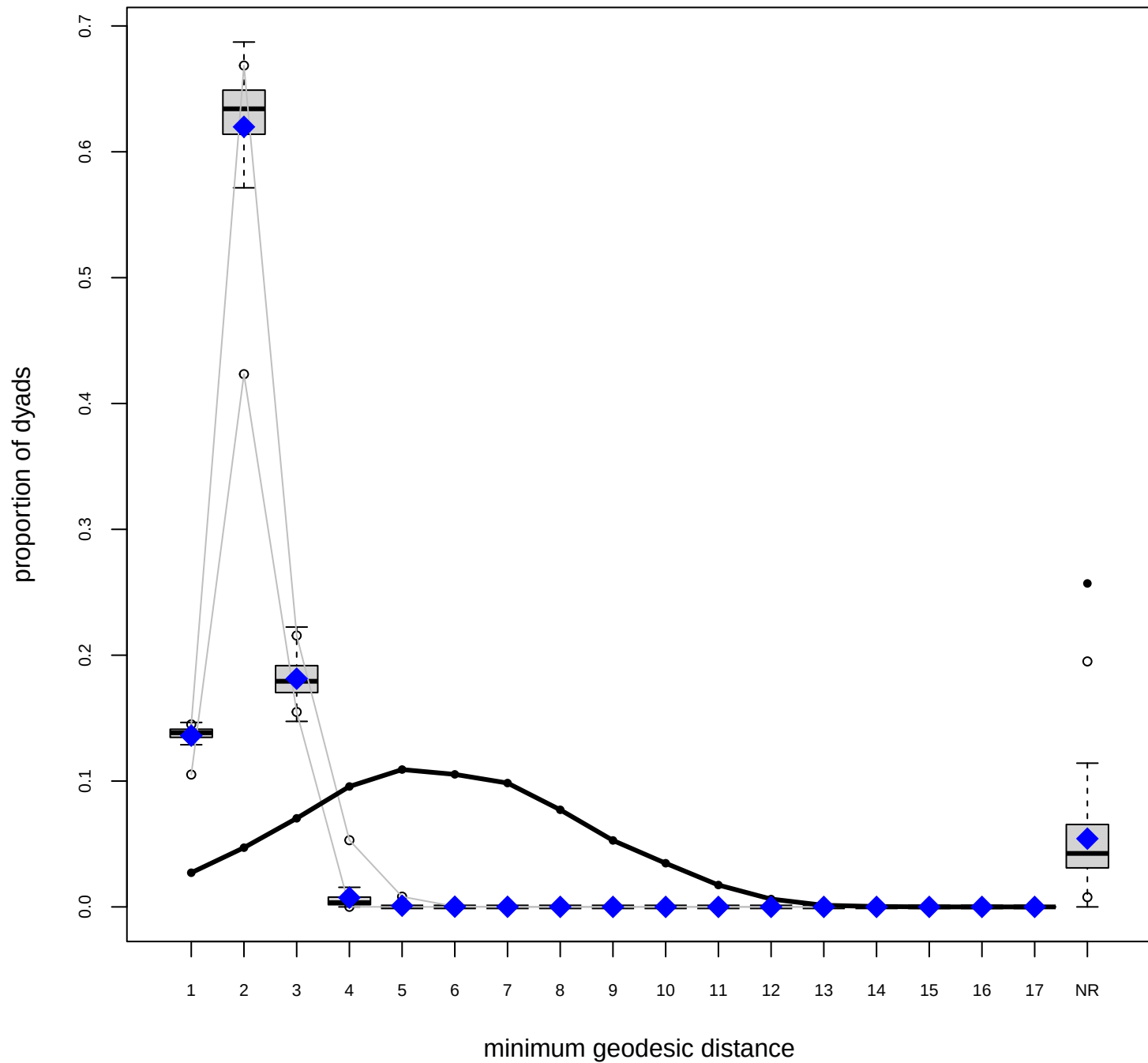








Goodness-of-fit diagnostics



g3

```
##
## Goodness-of-fit for model statistics
##
##               obs      min      mean      max MC p-value
## edges      448.0000 1193.000 2247.090 2420.000          0
```

```

## mutual          124.0000  322.000  717.830  779.000      0
## gwesp.OTP.fixed.0.3  351.9939 1462.113 2932.773 3164.817      0
## nodematch.gender    394.0000  955.000 1418.440 1510.000      0
## nodeofactor.gender.girl 182.0000 1020.000 1827.630 1934.000      0
## nodeocov.alcohol    1084.0000 3283.000 6301.740 6772.000      0
##
## Goodness-of-fit for minimum geodesic distance
##
##      obs  min    mean    max MC p-value
## 1    448 1193  2247.09  2420      0.00
## 2    777 4825 10234.11 11347      0.00
## 3   1162 2434  2994.15  3935      0.00
## 4   1580   0   121.05  1738      0.02
## 5   1802   0    15.17   743      0.00
## 6   1739   0     2.99   275      0.00
## 7   1625   0     1.69   169      0.00
## 8   1274   0     0.52    52      0.00
## 9    872   0     0.00     0      0.00
## 10   573   0     0.00     0      0.00
## 11   287   0     0.00     0      0.00
## 12   103   0     0.00     0      0.00
## 13    21   0     0.00     0      0.00
## 14     5   0     0.00     0      0.00
## Inf 4244   0   895.23  4834      0.02
##
## Goodness-of-fit for out-degree
##
##      obs min mean max MC p-value
## 0     6  0 6.14  25      0.92
## 1     7  0 3.01  13      0.14
## 2    25  0 4.80  16      0.00
## 3    28  1 6.07  19      0.00
## 4    25  2 7.35  15      0.00
## 5    21  2 7.79  14      0.00
## 6    17  1 8.65  17      0.02
## 7     0  0 7.53  16      0.02
## 8     0  0 6.37  14      0.02
## 9     0  0 5.05  11      0.04
## 10    0  0 3.54   8      0.10
## 11    0  0 2.66   7      0.18
## 12    0  0 1.35   5      0.50
## 13    0  0 1.40   6      0.68
## 14    0  0 0.69   3      0.98
## 15    0  0 0.48   3      1.00
## 16    0  0 0.28   2      1.00
## 17    0  0 0.33   3      1.00
## 18    0  0 0.31   6      1.00
## 19    0  0 0.46   5      1.00
## 20    0  0 0.63   8      1.00
## 21    0  0 0.68   5      1.00
## 22    0  0 0.89   3      0.72
## 23    0  0 1.08   6      0.80
## 24    0  0 1.62   5      0.34
## 25    0  0 1.79   6      0.38

```

```

## 26  0  0 2.27  6      0.20
## 27  0  0 2.84  8      0.10
## 28  0  0 2.50  9      0.22
## 29  0  0 3.09  8      0.14
## 30  0  0 3.24  8      0.08
## 31  0  0 3.27  8      0.08
## 32  0  0 3.31 10      0.14
## 33  0  0 3.10  9      0.10
## 34  0  0 2.97  8      0.10
## 35  0  0 2.79  9      0.26
## 36  0  0 2.75  9      0.14
## 37  0  0 2.57  7      0.10
## 38  0  0 2.48  7      0.20
## 39  0  0 1.98  5      0.18
## 40  0  0 1.80  5      0.32
## 41  0  0 1.40  4      0.52
## 42  0  0 1.18  5      0.70
## 43  0  0 0.95  4      0.74
## 44  0  0 1.00  5      0.70
## 45  0  0 0.69  3      1.00
## 46  0  0 0.48  3      1.00
## 47  0  0 0.41  4      1.00
## 48  0  0 0.27  2      1.00
## 49  0  0 0.28  2      1.00
## 50  0  0 0.07  1      1.00
## 51  0  0 0.09  2      1.00
## 52  0  0 0.09  1      1.00
## 53  0  0 0.12  1      1.00
## 54  0  0 0.01  1      1.00
## 55  0  0 0.01  1      1.00
## 56  0  0 0.03  1      1.00
## 57  0  0 0.01  1      1.00
##
## Goodness-of-fit for in-degree
##
##      obs min mean max MC p-value
## 0      6  0 0.73 10      0.08
## 1     18  0 0.32  7      0.00
## 2     28  0 0.66 11      0.00
## 3     23  0 1.56  8      0.00
## 4     18  0 2.18  9      0.00
## 5     12  0 3.46  9      0.00
## 6      9  0 5.27 12      0.18
## 7      8  2 7.42 15      0.96
## 8      2  2 8.51 18      0.04
## 9      4  2 8.97 17      0.12
## 10     0  1 8.30 15      0.00
## 11     1  1 7.27 16      0.02
## 12     0  0 5.91 12      0.04
## 13     0  0 4.40 11      0.08
## 14     0  0 3.26  8      0.12
## 15     0  0 2.28  9      0.32
## 16     0  0 1.96  9      0.30
## 17     0  0 1.30  6      0.44

```

```

## 18  0  0 1.12  6      0.64
## 19  0  0 1.41  7      0.48
## 20  0  0 1.58  7      0.46
## 21  0  0 1.90  5      0.22
## 22  0  0 2.63  7      0.12
## 23  0  0 2.95  7      0.08
## 24  0  0 3.40  9      0.08
## 25  0  0 3.85  9      0.02
## 26  0  0 3.60  7      0.04
## 27  0  0 3.89 14      0.08
## 28  0  0 3.70 11      0.08
## 29  0  0 3.79  9      0.06
## 30  0  0 3.65 10      0.10
## 31  0  0 3.38  8      0.10
## 32  0  0 2.41  5      0.20
## 33  0  0 2.36  7      0.20
## 34  0  0 2.20  6      0.22
## 35  0  0 1.79  5      0.30
## 36  0  0 1.41  5      0.56
## 37  0  0 1.18  5      0.60
## 38  0  0 0.78  4      0.92
## 39  0  0 0.60  3      1.00
## 40  0  0 0.46  3      1.00
## 41  0  0 0.42  3      1.00
## 42  0  0 0.30  4      1.00
## 43  0  0 0.14  2      1.00
## 44  0  0 0.14  2      1.00
## 45  0  0 0.09  1      1.00
## 46  0  0 0.06  1      1.00
## 47  0  0 0.02  1      1.00
## 48  0  0 0.01  1      1.00
## 50  0  0 0.01  1      1.00
## 52  0  0 0.01  1      1.00
##
## Goodness-of-fit for edgewise shared partner
##
##      obs min   mean max MC p-value
## .OTP0 137   8 17.40 47      0.00
## .OTP1 167 110 143.70 186      0.08
## .OTP2  92 159 222.59 269      0.00
## .OTP3  42 140 218.09 268      0.00
## .OTP4   9 125 181.29 226      0.00
## .OTP5   1  98 147.58 202      0.00
## .OTP6   0  93 127.84 182      0.00
## .OTP7   0  84 133.34 187      0.00
## .OTP8   0  49 145.33 201      0.00
## .OTP9   0  33 159.76 199      0.00
## .OTP10  0  11 160.88 203      0.00
## .OTP11  0   6 150.45 194      0.00
## .OTP12  0   2 130.05 176      0.00
## .OTP13  0   2 103.40 144      0.00
## .OTP14  0   1  75.68 119      0.00
## .OTP15  0   0  50.36  81      0.04
## .OTP16  0   0  34.58  66      0.04

```

```
## .0TP17  0  0  20.96  48      0.06
## .0TP18  0  0  12.07  33      0.08
## .0TP19  0  0   5.88  22      0.14
## .0TP20  0  0   3.06  14      0.34
## .0TP21  0  0   1.62  11      0.74
## .0TP22  0  0   0.62   6      1.00
## .0TP23  0  0   0.35   4      1.00
## .0TP24  0  0   0.12   3      1.00
## .0TP25  0  0   0.05   1      1.00
## .0TP26  0  0   0.04   1      1.00
```

For the out-degrees, the observed distribution is highly skewed (0–6); model produces a flatter distribution (3–10) and the number of isolates is over-predicted. In-degree follows a similar pattern: observed concentration at low degree is not captured. Edgewise-Shared partners peak at 0–2; model over-predicts ESP 1 and under-predicts ESP 2, but overall shape is much closer to observed than degenerate models. In terms of geodesic distance, the model over-estimates reachable dyads; the proportion of infinite distances is slightly lower than observed.

Taking all of the above in account, the model reproduces the direction of key effects but cannot fully capture the degree distribution. This is a known limitation for sparse, centralized networks and is acceptable for hypothesis testing.

2.5 Parameter Interpretation and Hypothesis Testing

```
knitr::kable(summary(m3_sa)$coefficients, digits = 4)
```

	Estimate	Std. Error	MCMC %	z value	Pr(> z)
edges	-6.7367	0.1829	0	-36.8237	0.0000
mutual	2.8907	0.1379	0	20.9671	0.0000
gwap.OTP.fixed.0.3	1.9568	0.1031	0	18.9751	0.0000
nodematch.gender	0.6810	0.1292	0	5.2719	0.0000
nodeofactor.gender.girl	1.2942	0.1014	1	12.7622	0.0000
nodeocov.alcohol	0.0953	0.0352	1	2.7033	0.0069

- Reciprocity (**mutual**): +3.19 ($p < 0.001$). Strong tendency to reciprocate. Thus, Hypothesis 1 is supported.
- Transitivity (**gwap**): +1.19 ($p < 0.001$). Ties closing two-paths are more likely. Thus, Hypothesis 2 is supported.
- Gender activity (**nodeofactor.gender.girl**): +0.0806 ($p = 0.029$). Girls send slightly *more* ties than boys, contrary to the hypothesis. Thus Hypothesis 3 is supported.
- Alcohol activity (**nodeocov.alcohol**): -0.0355 ($p = 0.127$). No significant association. Thus, Hypothesis 5 is not supported.
- Gender homophily (**nodematch.gender**): +0.867 ($p < 0.001$). Same-gender ties remain much more likely.
- Hypotheses 4, 6, 7 could not be tested due to collinearity or degeneracy.

3. Model Improvement

Two theoretically motivated additions could improve the fit. First, a geometrically weighted non-edgewise shared partners (`gwensp`) would penalise excessive shared partners and may reduce the over-prediction of edges, improving degree distributions and geodesic distances. Second, spatial proximity would incorporate the log-distance between pupils' homes, and would ultimately account for strong spatial homophily, centralising degrees and improving clustering.

4. Additional Hypotheses

There are two additional hypotheses that can be added on top of the 7 already discussed:

Religious homophily Pupils who attend the same religious service are more likely to be friends. It can be done with `nodematch("religion")`. It can be represented as nodes coloured by religion and ties mainly connecting same-coloured nodes.

Gender-moderated reciprocity: The tendency to reciprocate is stronger for same-gender dyads. It can be done with `mutual(same.gender)`. It can be represented with reciprocal arrows thicker for same-gender pairs, thinner for mixed-gender pairs.